

## Task 1: Five Main Phases of Ethical Hacking

| Phase                    | Goal                                                                                                                 |
|--------------------------|----------------------------------------------------------------------------------------------------------------------|
| Reconnaissance           | Collect information about the target.                                                                                |
| Scanning                 | Identify hosts, ports, services and possible vulnerabilities.                                                        |
| Gaining Access           | Exploit an authorized vulnerability to demonstrate security weaknesses.                                              |
| Maintaining Access       | In an authorized test, determine whether access could persist without detection.                                     |
| Analysis/Covering Tracks | Assess evidence, detection capability and security impact; ethical testers document rather than hide their activity. |

## Task 2: Instagram Account Compromise — Digital Evidence

If an Instagram account appears compromised, three useful categories of evidence include:

1. Device/Application Evidence
  - Instagram application data
  - Login/session information
  - Application notifications
  - Relevant timestamps
2. Network Evidence
  - IP addresses
  - DNS/network logs
  - Connection timestamps
  - Router/firewall logs where available
3. Account/Communication Evidence
  - Security emails

- Login alerts
- Password-reset notifications
- Suspicious messages or posts
- Account activity history

Evidence should be preserved carefully so that it is not accidentally modified or deleted.

### Task 3: Windows Event ID 4624

#### Objective

Find the last successful login before suspicious activity.

#### Steps

1. Open the provided Windows event log.
2. Search for:
  - 4624
3. Event ID 4624 represents a successful logon.
4. Check the:
  - Date
  - Time
  - Account/username
  - Logon type
5. Compare the timestamps with the suspicious activity.
6. Identify the most recent successful login before the suspicious event.

#### Report Format

| Field      | Finding                               |
|------------|---------------------------------------|
| Event ID   | 4624                                  |
| Timestamp  | [Insert timestamp from trainer's log] |
| Username   | [Insert username]                     |
| Logon Type | [Insert value]                        |

Significance      Successful authentication before suspicious activity

Important: You need the actual trainer-provided log file to fill these values.  
Don't invent them.

#### Task 4: zomato\_log.txt Forensic Analysis

##### Sample Log

Create a file named:

zomato\_log.txt

Example contents:

```
2026-10-05 09:00:01 username=nishit status=FAILED ip=192.168.1.10
2026-10-05 09:02:15 username=nishit status=SUCCESS ip=192.168.1.10
2026-10-05 09:05:32 username=rahul status=FAILED ip=192.168.1.20
2026-10-05 09:07:11 username=rahul status=SUCCESS ip=192.168.1.20
2026-10-05 09:10:45 username=admin status=FAILED ip=192.168.1.30
2026-10-05 09:11:20 username=admin status=FAILED ip=192.168.1.30
2026-10-05 09:12:04 username=admin status=SUCCESS ip=192.168.1.30
2026-10-05 09:15:55 username=amit status=FAILED ip=192.168.1.40
2026-10-05 09:17:22 username=amit status=SUCCESS ip=192.168.1.40
2026-10-05 09:20:31 username=test status=FAILED ip=192.168.1.50
```

##### Analysis Steps

1. Open zomato\_log.txt.
2. Examine each login entry.
3. Identify the status field.
4. Mark:
  - SUCCESS = successful login
  - FAILED = unsuccessful login
5. Record the username, IP address and timestamp.
6. Look for repeated failed attempts.

7. Check whether a successful login occurred after repeated failures.
8. Summarize suspicious patterns.

## Findings

| <b>Username</b> | <b>Successful</b> | <b>Failed</b> | <b>Observation</b>                    |
|-----------------|-------------------|---------------|---------------------------------------|
| nishit          | 1                 | 1             | Successful login after one failure    |
| rahul           | 1                 | 1             | Successful login after one failure    |
| admin           | 1                 | 2             | Multiple failures followed by success |
| amit            | 1                 | 1             | Successful login after one failure    |
| test            | 0                 | 1             | No successful login                   |

## Suspicious Activity

The admin account is the most interesting entry because it had multiple failed login attempts followed by a successful login. A forensic investigator should examine the associated IP address, timestamps and other authentication evidence to determine whether the successful login was legitimate.

## Task 5: Digital Forensic Analysis Checklist

### Basic Forensic Checklist

1. Identify the incident and define the scope of investigation.
2. Preserve evidence to prevent accidental modification.
3. Document the system state and relevant information.
4. Acquire forensic copies of relevant storage or data where authorized.
5. Collect logs and authentication records from relevant systems.

6. Analyze timestamps and user activity to establish a timeline.
7. Identify suspicious files, processes, accounts and network activity.
8. Correlate evidence from multiple sources.
9. Document findings and maintain chain of custody.
10. Prepare a final forensic report containing evidence, analysis and conclusions.

#### My Additional Step for a Social Media Account Hack

I would add checking account security notifications and login/session history because social media compromises often leave evidence such as unfamiliar login locations, devices, password-reset requests or suspicious session activity.